nature nanotechnology

Article

https://doi.org/10.1038/s41565-025-01963-8

Nonlinear Nernst effect in trilayer graphene at zero magnetic field

Received: 16 October 2024

Accepted: 15 May 2025

Published online: 23 June 2025

Check for updates

Hao Liu[1,9], Jingru Li[1,9], Zhifan Zhang[2,9], Jinfeng Zhai[1], Min Zhang[1], Hua Jiang[1,2] ✉, X. C. Xie[2,3,4], Pan He[1,4] ✉ & Jian Shen[1,4,5,6,7,8] ✉

The Nernst effect, that is, the generation of a transverse voltage in response to a temperature gradient, enables thermoelectric energy conversion. In the absence of an external magnetic field, the linear Nernst effect is forbidden in non-magnetic materials because of time-reversal symmetry constraints, but the recently predicted nonlinear Nernst effect (NNE) is allowed. Here we report the experimental observation of the NNE in non-magnetic ABA trilayer graphene, even in the absence of an external magnetic field. This effect is detected via electric harmonic measurements under an alternating temperature gradient at temperatures below 12 K. The NNE exhibits a quadratic dependence on the temperature gradient. It is notably enhanced near the charge neutrality point and reaches a giant effective Nernst coefficient of up to 300 μV K^{-1} at 2 K, surpassing the linear coefficients of magnetic materials. Moreover, we establish a scaling law between the NNE and the linear Seebeck effect, confirming the dominance of a skew scattering mechanism in driving the NNE. Our findings demonstrate an alternative approach for thermoelectric energy harvesting and cooling applications via nonlinear thermoelectric responses, which may, in the long run, offer alternative approaches towards the development of advanced thermoelectric devices.

The Nernst effect, a thermoelectric phenomenon characterized by the generation of a transverse voltage in response to a longitudinal temperature gradient ∇T_x (ref. 1), has garnered substantial attention for its potential applications in energy harvesting and cooling[2,3]. The conventional Nernst effect requires time-reversal symmetry breaking (Fig. 1a). For example, the ordinary Nernst effect occurs under a perpendicular magnetic field. The magnetic field dependence poses an important challenge to thermoelectric device miniaturization and circuit integration. This limitation can be addressed by exploring the anomalous Nernst effect in magnetic materials. Large anomalous Nernst coefficients S_{yx} surpassing 5 μV K^{-1} were demonstrated recently in a few magnetic materials[2–7]. Despite these advancements, a magnetic field is still required to align the magnetization perpendicular to the ∇T_x owing to the soft ferromagnetic properties of these materials. The highest S_{yx} achieved without a magnetic field so far is about 5.6 μV K^{-1}, utilizing hard ferromagnets such as Co_3Sn_2S (refs. 8,9), Fe_3Pt (ref. 10) and $SmCo_5$ (refs. 11,12). It is crucial to further enhance S_{yx} for practical thermoelectric applications, a challenge that is being addressed through the design of band structures[5] with large Berry curvatures and the discovery of new working mechanisms[13].

These previous studies of the Nernst effect have focused primarily on linear response, where the generated voltage is directly proportional to the temperature gradient. These linear responses typically necessitate the presence of magnetic fields or magnetic orderings to break time-reversal symmetry. It is noteworthy that the Nernst effect and the Hall effect are two analogous effects related by the Mott relation. While the linear Hall effect requires the breaking of time-reversal

[1]State Key Laboratory of Surface Physics, Institute for Nanoelectronic Devices and Quantum Computing, Fudan University, Shanghai, China. [2]Interdisciplinary Center for Theoretical Physics and Information Sciences, Fudan University, Shanghai, China. [3]International Center for Quantum Materials, School of Physics, Peking University, Beijing, China. [4]Hefei National Laboratory, Hefei, China. [5]Department of Physics, Fudan University, Shanghai, China. [6]Shanghai Research Center for Quantum Sciences, Shanghai, China. [7]Zhangjiang Fudan International Innovation Center, Fudan University, Shanghai, China. [8]Collaborative Innovation Center of Advanced Microstructures, Nanjing, China. [9]These authors contributed equally: Hao Liu, Jingru Li, Zhifan Zhang. ✉e-mail: jianghuaphy@fudan.edu.cn; hepan@fudan.edu.cn; shenj5494@fudan.edu.cn

https://doi.org/10.1038/s41565-025-01963-8

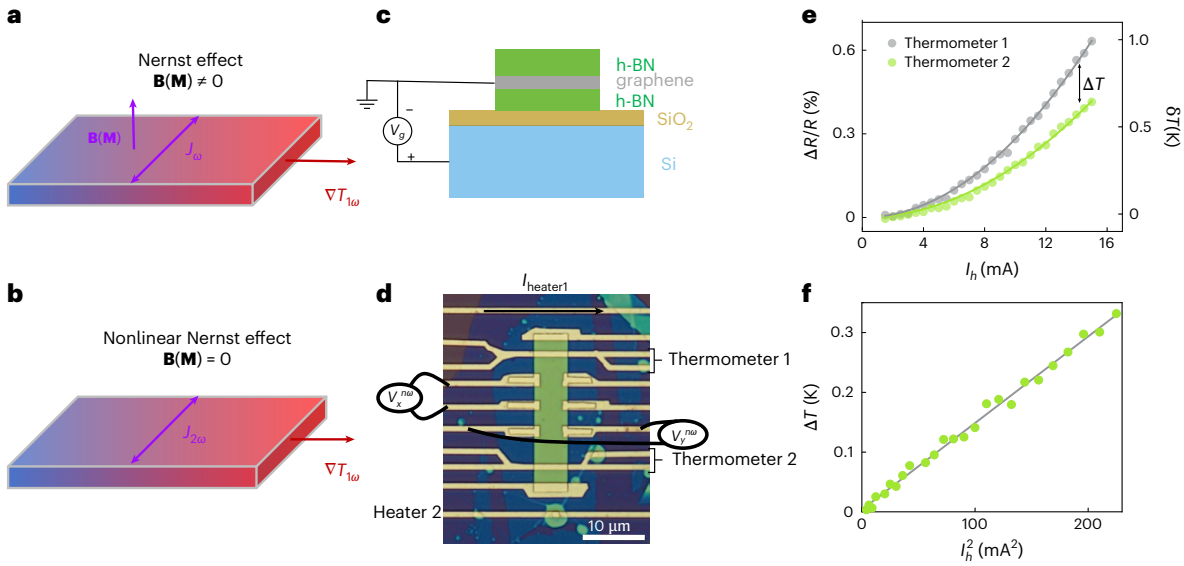


Fig. 1 | Schematic illustration of NNE and trilayer graphene device for measuring NNE. a,b, Diagrams depicting the thermal-gradient-driven linear Nernst effect under a magnetic field (**B**) or magnetic order (**M**) (**a**) and NNE in the absence of **B** and **M** (**b**). **c**, A side view of the van der Waals heterostructure used for thermoelectrical measurements, showing that the trilayer graphene is encapsulated by two h-BN. A back gate voltage (V_g) is applied between the trilayer graphene and a highly doped silicon substrate to modulate the graphene carrier density. **d**, An optical image of a typical device used for thermoelectrical and electric measurements. Scale bar, 10 μm. Two sets of four-terminal electrodes function as thermometers (thermometer 1 and thermometer 2) to monitor the local temperatures by measuring their four-probe resistances. In addition, the two heaters, disconnected from the graphene, apply opposite temperature fluctuations through the sample under an a.c. current. Three pairs of Hall electrodes are configured for detecting harmonic voltages under an a.c. temperature gradient or current. **e**, The normalized resistance changes ($\Delta R/R$) of the two thermometers as a function of heater current I_h at 20 K. The changes in resistance are mapped to the changes in temperature δT at the thermometers. ΔT is defined as the temperature difference between two thermometers. **f**, ΔT scales linearly with the square of I_h, indicative of Joule heating as the underlying mechanism. The length and width of the device is 4.6 μm and 6 μm, respectively. The distance between two thermometers is 20 μm in our sample.

symmetry, the second-order nonlinear Hall effect (NHE) is not subject to such a constraint[14]. The recent discovery of NHE in non-centrosymmetric materials[15–17] has opened up new avenues for exploring nonlinear thermoelectric phenomena utilizing the quantum geometry of electron wave function and impurity scatterings[14,18–22]. However, the nonlinear Nernst effect (NNE), analogous to NHE but driven by a thermal gradient (Fig. 1b), has been much less explored.

The NNE has only recently been predicted in theory. Intrinsic NNE has been predicted to arise from the orbital magnetic quadrupole moment[23]. In addition, the NNE emerging from a pseudotensorial quantity[24] similar to a Berry curvature dipole (BCD) was predicted in non-centrosymmetric systems[24–26], such as bilayer WTe_2 (ref. 24), strained transition-metal dichalcogenides[25] and strained graphene[26]. Furthermore, the disorder-mediated scatterings have been predicted to give rise to extrinsic NNEs[27]. Despite its theoretical promise and potential practical implications, experimental evidence of the NNE has remained elusive. Thus, demonstrating the NNE and validating these theoretical predictions are of paramount importance.

In this Article, we report the experimental observation of the NNE in non-magnetic ABA trilayer graphene at temperatures below 12 K and in the absence of a magnetic field. This material represents the thinnest pristine graphene that breaks inversion symmetry while preserving time-reversal symmetry. Using low-frequency electric harmonic measurements under an alternating thermal gradient[28], we successfully detected and characterized the NNE in ABA trilayer graphene. Leveraging the ultrahigh mobility of graphene, we achieved a Nernst coefficient of up to 300 μV K^{-1} (ref. 17). In addition, we established a scaling law between the NNE and the linear Seebeck effect. Based on this scaling law, symmetry analysis and theoretical study, we identify skew scattering as the dominant mechanism driving the NNE. Our findings offer key insights into the underlying mechanisms of the NNE and open new avenues for advancing thermoelectric devices.

Results

To fabricate high-quality ABA trilayer graphene encapsulated by hexagonal boron nitride (h-BN), the trilayer graphene and h-BN flakes were first prepared by mechanical exfoliation on Si substrates. After selection under a microscope, the graphene and h-BN layers were sequentially stacked into a sandwich structure using the pick-up method and transferred onto a highly doped Si substrate with a 285-nm SiO_2 layer (Fig. 1c). The Hall bar device was patterned using the electron beam lithography and reactive ion etching, followed by the deposition of metallic electrodes Cr and Au (3 nm and 90 nm, respectively) through thermal evaporation. In addition to the electrodes for all-electric measurements, the electrodes for thermoelectric measurements were also designed, as shown in Fig. 1d. A controllable temperature gradient ∇T can be applied to the sample by passing a current I_h through a microfabricated heater. The ∇T can be determined by measuring the temperature difference ΔT between the thermometers 1 and 2. The temperature at each thermometer was determined by its temperature-dependent four-point resistance[29]. The length of the heater (160 μm) was much larger than the Hall bar width (15 μm) to ensure a uniform and longitudinal temperature gradient along the channel[30]. As shown in Fig. 1e, the temperature variations δT at the two thermometers with the heater current were evaluated by measuring their resistance changes (Methods). Figure 1f shows that the resulting ΔT scales linearly with I_h^2 owing to the Joule heating effect. The range of I_h was restricted to satisfy that ΔT was much smaller than T, ensuring that experiments were performed in the linear heat transport regime[29].

Observation of NNE at zero magnetic field

In our experiment, a sinusoidal current $I_h = I\sin\omega t$, where t is time, with a frequency of ω (17.777 Hz) was applied to the heater, inducing a ΔT with a frequency of 2ω. The longitudinal Seebeck effect and the

https://doi.org/10.1038/s41565-025-01963-8

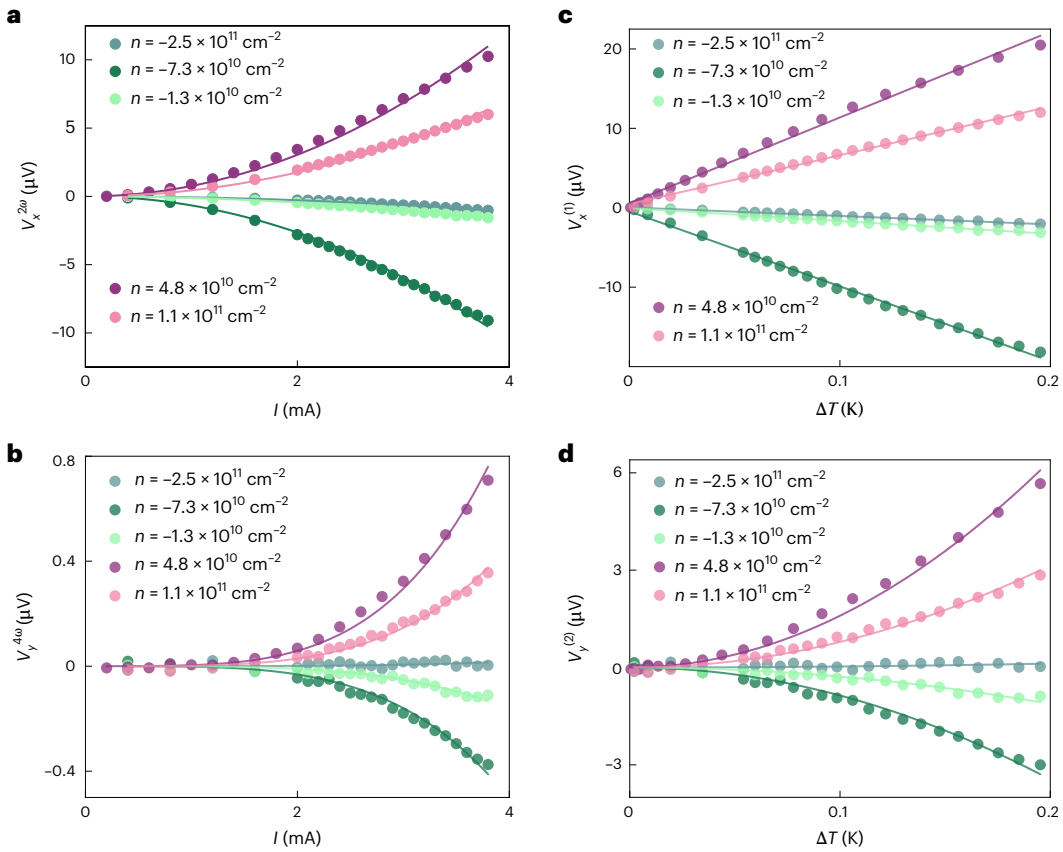


Fig. 2 | Measurements of Seebeck effect and NNE in trilayer graphene. a,b, The second-harmonic voltage $V_x^{2\omega}$ (**a**) parallel to the temperature gradient and the fourth-harmonic voltage $V_y^{4\omega}$ (**b**) transverse to the temperature gradient as functions of the a.c. heating current amplitude I for a few carrier densities n. The data are fitted with quadratic (**a**) and fourth power (**b**) relations, respectively. **c**,**d**, The Seebeck effect voltage $V_x^{(1)}$ (**c**) and the NNE voltage $V_y^{(2)}$ (**d**) as a function of the temperature difference ΔT between thermometer 1 and thermometer 2. These data are well fitted by linear (**c**) and quadratic (**d**) curves, respectively. The $V_x^{(1)}$ $(V_y^{(2)})$ is linearly proportional to the $V_x^{2\omega}$ $(V_y^{4\omega})$. ΔT is converted from I. These data were measured at 5 K and zero magnetic field.

transverse Nernst effect in response to ΔT was probed by harmonic measurements using lock-in amplifiers at zero magnetic field. For the Seebeck effect with a linear relationship with ΔT, the second-harmonic longitudinal voltage $V_x^{2\omega}$ appears and scales quadratically with the heating current amplitude I, as shown in Fig. 2a. To probe the NNE exhibiting a second-order response to ΔT, the fourth-harmonic transverse voltage $V_y^{4\omega}$ was measured. Remarkably, we observed a sizable $V_y^{4\omega}$, which displays a fourth power relation with I in Fig. 2b. We also measured 1ω and 3ω signals (shown in Extended Data Fig. 1), which are negligible compared with the 4ω response, consistent with the expectation of thermoelectric effects occurring at multiples of 2ω. By converting I into ΔT, the Seebeck voltage $V_x^{(1)}$ and the NNE voltage $V_y^{(2)}$ were plotted as a function of ΔT in Fig. 2c and Fig. 2d, respectively. $V_x^{(1)} = 2V_x^{2\omega}$ $\left(V_y^{(2)} = 8V_y^{4\omega}\right)$ is derived as detailed in Supplementary Note 1. Good linear and quadratic fittings to the thermoelectric data are shown in Fig. 2c,d for various carrier densities, which agree well with the characteristics of Seebeck effect and NNE, respectively. It is worth noting that the magnitude of $V_y^{(2)}$ (Fig. 2d) is comparable to the value of $V_x^{(1)}$ (Fig. 2c). The NNE observed in this work provides a new approach for transverse thermoelectric device, which has several advantages over the longitudinal counterparts, such as simple lateral structure, convenient scalability and easy fabrication. In addition, transverse thermoelectric devices avoid contact problems encountered in thermoelectric modules based on the Seebeck effect[2,3]. Notably, the Nernst coefficient in ABA graphene can be two orders of magnitude larger than the highest value found in magnetic materials at zero magnetic field[8,10]. Furthermore, the NNE-based devices do not require an external magnetic field or magnetic materials, and generate no stray field, making them promising for thermoelectric applications.

To further confirm the observation of NNE, we applied an opposite temperature gradient ∇T to the sample by using the heater at the other side (Fig. 1d). Schematics of the measurement setups for the opposite ∇T are shown in Fig. 3a for a comparison. The thermoelectrical signals as a function of the carrier density n are plotted in Fig. 3b,c for the two cases. While the electrical field generated from Seebeck effect $E_x^{(1)}$ changes sign when the temperature gradient is reversed, the electrical field $E_y^{(2)}$ generated from NNE remains the same sign at all carrier densities. The $E_x^{(1)}$ $\left(E_y^{(2)}\right)$ is calculated from the voltage drop $V_x^{(1)}$ $\left(V_y^{(2)}\right)$ divided by the length L (width W) of device, that is, $E_x^{(1)} = \frac{V_x^{(1)}}{L}$, $E_y^{(2)} = \frac{V_y^{(2)}}{W}$. The change (no change) in the sign of $E_x^{(1)}$ $\left(E_y^{(2)}\right)$ under opposite ∇T is consistent with the nature of linear (second-order nonlinear) thermoelectric effect. The NNE was also found to be independent of the a.c. current frequency, as shown in Extended Data Fig. 2. Moreover, we excluded nonlinear transport signals from gate voltage oscillations, as recently reported for $MnBi_2Te_4$ (ref. 31). This is demonstrated in Extended Data Fig. 3. These experimental results unambiguously confirm the observation of NNE, which exists in ABA trilayer graphene without the need for an external magnetic field. The NNE broadens the material platform for generating spontaneous transverse thermoelectric effects.

The Mott relation between NNE and NHE

The $E_x^{(1)}$ reverses its sign when the Fermi level moves across the charge neutral point (CNP) in Fig. 3b, because the carrier changes from electron

https://doi.org/10.1038/s41565-025-01963-8

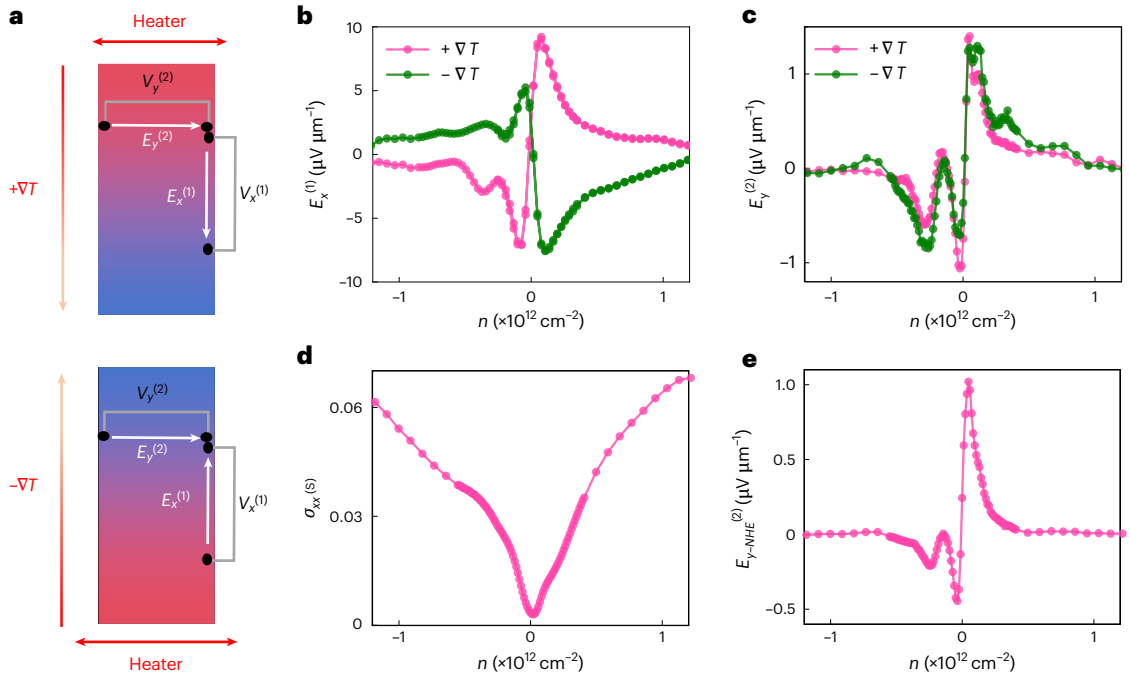


Fig. 3 | The carrier density dependences of thermoelectric effects and NHE at T = 5 K. a, The configurations for thermoelectrical measurement under two opposite ∇T **b**,**c**, The carrier density dependences of the electric field generated from Seebeck effect $E_x^{(1)}$ (**b**) and the NNE $E_y^{(2)}$ (**c**) for the two reversed temperature gradient directions. The NNE is an even function of ∇T, leading to the same sign of $E_y^{(2)}$ generated from the NNE when the ∇T is reversed in **c**. By contrast, the $E_x^{(1)}$ generated from Seebeck effect is an odd function of ∇T, which thus changes sign in **b**. The heating current is 4 mA, and the temperature difference between two thermometers is 0.22 K. **d**,**e**, The longitudinal conductivity σ_{xx} (**d**) and the NHE generated electric field $E_{y-\mathrm{NHE}}^{(2)}$ (**e**) as functions of carrier density n.

to hole[32]. Similarly, the $E_y^{(2)}$ shows a sign reversal at CNP in Fig. 3c, with peaks and dips around the CNP. To better understand the NNE results, we also conducted electric-driven transport measurements by applying a current along the longitudinal direction. The measured longitudinal conductivity σ_{xx} and NHE are shown in Fig. 3d,e. The σ_{xx} reaches a minimum at CNP and increases rapidly for larger values of $|n|$. Notably, the electric field $E_{y-\mathrm{NHE}}^{(2)}$ generated by the NHE in Fig. 3e shows a carrier density dependence similar to that of the NNE in Fig. 3c, suggesting a common origin for these two nonlinear transport effects, despite being driven by different forces.

The Mott relation predicts that the thermoelectric coefficient is proportional to the derivative of the electrical conductivity with respect to the chemical potential[33]. The validity of this relation has been tested for the longitudinal linear transports (Supplementary Note 2). For the transverse transports, both the linear Hall effect and the Nernst effect vanish under zero magnetic field in trilayer graphene. Nevertheless, the NHE and the NNE have been demonstrated in this work in the absence of magnetic field. A fundamental question is whether there is a Mott relation between the nonlinear electric and the nonlinear thermoelectric coefficients. Recent theoretical works have established a Mott formula in the nonlinear regime[34–36] that shows that the NHE and the NNE conductivities are linearly proportional to each other. However, the experimental verification of the Mott relation is still missing. Our experimental data shown in Fig. 3c,e indicate that the variations of the NHE and the NNE with carrier density are similar, indicating the linear relationship between the NHE and the NNE conductivities (Extended Data Fig. 4 and Supplementary Fig. 1).

The scaling law of NNE and its dominant physical origin

To uncover the physical origin of the NNE, we measured the thermoelectric signals at different temperatures. The Seebeck coefficient S_{xx} and the NNE coefficient $S_{yxx}^{(2)}$ as a function of n obtained at various temperatures below 12 K are shown in Fig. 4a and Fig. 4b, respectively. The S_{xx} and $S_{yxx}^{(2)}$ are defined as $S_{xx} = \frac{E_x^{(1)}}{\nabla T}$ and $S_{yxx}^{(2)} = \frac{E_y^{(2)}}{(\nabla T)^2}$, respectively. The temperature gradient ∇T is defined as $\nabla T = \frac{\Delta T}{L_{12}}$, where L_{12} is the distance between the two thermometers. Both thermoelectrical coefficients decrease rapidly with rising temperature. The S_{xx} and $S_{yxx}^{(2)}$ values at two representative peaks are plotted as a function of temperature in Fig. 4c and Fig. 4d, respectively. The $S_{yxx}^{(2)}$ decreases more rapidly than S_{xx} as the temperature increases, and $S_{yxx}^{(2)}$ diminishes when the temperature reaches 12 K. In the following, we shall explore the relation between these two thermoelectrical coefficients and draw comparisons with their electrical counterparts.

It should be noted that the NNE and the NHE present two analogous nonlinear transports driven by the thermal gradient ∇T_x and electric field E_x, respectively. With the nonlinear Nernst conductivity $\alpha_{yxx}^{(2)}$ and nonlinear Hall conductivity $\sigma_{yxx}^{(2)}$ (refs. 14,26), the transverse nonlinear current $J_y^{(2)}$ resulting from the NNE and the NHE can be expressed as follows:

$$J_y^{(2)} = \alpha_{yxx}^{(2)}(\nabla T_x)^2; \alpha_{yxx}^{(2)} = S_{yxx}^{(2)}\sigma_{xx} \quad (1)$$

$$J_{y-\mathrm{NHE}}^{(2)} = \sigma_{yxx}^{(2)}(E_x)^2; \sigma_{yxx}^{(2)} = \frac{E_{y-NHE}^{(2)}}{E_x^2}\sigma_{xx}. \quad (2)$$

These equations show that the $S_{yxx}^{(2)}$ in NNE corresponds to the $\frac{E_y^{(2)}}{E_x^2}$ in NHE. Meanwhile, Ohm's law and the Seebeck effect are similar linear longitudinal transports driven by the electric field and thermal gradient, respectively. The electric conductivity σ_{xx} is defined as the ratio of the current density to the applied electrical field, expressed as $\sigma_{xx} = \frac{J_x}{E_x}$. Thus, $S_{xx}\sigma_{xx} = \frac{J_x}{\nabla T}$ in the Seebeck effect corresponds to σ_{xx} in the Ohm's law. It is worth noting that the linear scaling law

https://doi.org/10.1038/s41565-025-01963-8

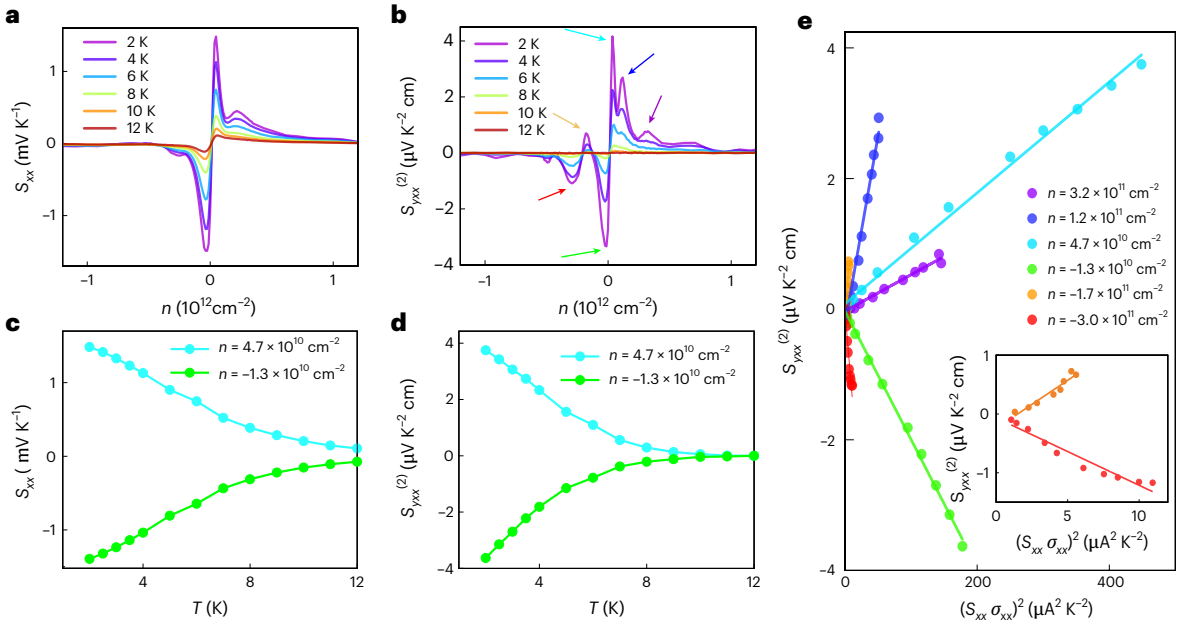


Fig. 4 | Temperature dependences of S_{xx} and $S_{yxx}^{(2)}$ and their scaling relation. **a**,**b**, Carrier density dependences of the S_{xx} (**a**) and the $S_{yxx}^{(2)}$ (**b**) at different temperatures. The $S_{yxx}^{(2)}$ exhibits several peaks near the charge neutrality point. **c**,**d**, The S_{xx} (**c**) and $S_{yxx}^{(2)}$ (**d**) as a function of temperature at two representative n, which are located at the main peaks in **b** as indicated by the light-blue and green arrows. **e**, The $S_{yxx}^{(2)}$ plotted as a function of $(S_{xx}\sigma_{xx})^2$ for different n located at the peaks in **b**. Inset: replots of the data at two carrier densities for clarity. The linear fits to the experimental data show nearly zero intercepts. The heating current is 4 mA, and the temperature difference between two thermometers ranges from 0.18 K to 0.25 K for the temperature region shown in the figure.

between $\frac{E_{y-\mathrm{NHE}}^{(2)}}{E_x^2}$ and $(\sigma_{xx})^2$, that is, $\frac{E_{y-\mathrm{NHE}}^{(2)}}{E_x^2} = a(\sigma_{xx})^2 + b$, has been widely used to study the physical origins of NHE[17,18,20]. The slope a and the intercept b represent contributions from skew scattering and BCD and/or side jump, respectively. However, the scaling law has not been discussed in NNE. Following the well-established scaling of NHE, we use the linear scaling law between $S_{yxx}^{(2)}$ and $(S_{xx}\sigma_{xx})^2$, that is, $S_{yxx}^{(2)} = A(S_{xx}\sigma_{xx})^2 + B$, to analyse the mechanisms of NNE. This scaling law has been derived theoretically on the basis of the dependence of NNE conductivity on relaxation time in Supplementary Note 3. The $S_{yxx}^{(2)}$ is plotted as a function of $(S_{xx}\sigma_{xx})^2$ for several typical carrier densities in Fig. 4e. As a result, a good linear relation can be found for all the data. From the linear fittings, we obtain a sizable slop A and a small intercept B for each carrier density, which may indicate that skew scattering dominates the NNE in the ABA trilayer graphene system. This result is consistent with the theoretical analysis: The C_3 symmetry of trilayer graphene excludes the contribution from BCD. Meanwhile, the two side-jump contributions from side-jump velocity and the anomalous distribution function cancel each other out[27] (Supplementary Note 3). Although additional side-jump contributions may exist[37,38], our experimental observations indicate that the dominant contribution to NNE in trilayer graphene arises from skew scattering. The graphene sample with an ultrahigh mobility of ~6×10^5 cm^2 Vs^{-1} enables a large skew scattering contribution in this work[17,20,39], as this contribution exhibits a high-order dependence on the relaxation time (Supplementary Note 3). We also applied alternative scaling laws[21,40] to fit our experimental data in Supplementary Note 4, which further confirms the dominant contribution from skew scattering.

We have reproduced the above results in three additional samples, including the ∇T-reversal-even characteristic of NNE and the scaling law of NNE, as shown in Supplementary Figs. 2–5. Our work provides experimental evidence and theoretical analysis demonstrating that skew scattering plays a crucial role in the NNE, leading to a giant effective Nernst coefficient $\left(S_{yx} = E_y^{(2)}/\nabla T\right)$ in graphene at zero magnetic field. The S_{yx} results from three different devices are summarized in Supplementary Fig. 6. A comparison of the S_{yx} among different materials[4–12] is shown in Extended Data Fig. 5. The S_{yx} in the ABA graphene can be around two orders of magnitude larger than the highest value observed in magnetic materials ($SmCo_5$ and $SmCo_5$-type[11,12]) at zero magnetic. It is one order of magnitude larger than the highest value in magnetic $UCo_{0.8}Ru_{0.2}Al$ under a magnetic field. Although the observed S_{yx} reaches a remarkably high value in NNE, a dedicated figure of merit must be further established to evaluate the efficiency of nonlinear thermoelectric devices (Supplementary Note 5). Moreover, the development of large-scale growth[41] and efficient mass transfer techniques for two-dimensional (2D) materials[42] is crucial for advancing 2D materials towards thermoelectric applications.

The second-order NNE is allowed in a wide variety of time-reversal-invariant but inversion-symmetry-breaking materials. We conducted control measurements on monolayer and bilayer graphene, which are inversion-symmetric in their pristine forms. We observed negligibly small nonlinear Nernst coefficients in these samples (Supplementary Note 6), in stark contrast to the pronounced response in trilayer graphene. These results are consistent with the symmetry requirements for second-order nonlinear effects and further reinforce our conclusions. Although the NNE is currently observed at low temperatures (below 12 K) in trilayer graphene, this effect holds the potential to be extended to other non-centrosymmetric material systems, where it may be realized at room temperature for broader practical applications. Although most real-world applications prioritize room-temperature thermoelectric materials, those operating at low temperatures remain valuable for specialized applications. Achieving room-temperature NNE requires specific physical properties and electronic structures (Supplementary Note 7). Essential criteria include broken inversion symmetry, as found in polar systems and chiral crystals, and strong Berry curvature near the Fermi energy, as observed in Weyl and Dirac semimetals. High carrier mobilities are also critical for efficient nonlinear thermoelectric transport. Furthermore, electronic structures

that support nonlinear responses over a broad range of Fermi levels are desirable. Notably, materials exhibiting a room-temperature NHE, such as Weyl semimetal $TaIrTe_4$ (ref. 16), Dirac semimetal $BaMnSb_2$ (ref. 43) and small-gap Weyl semiconductor Te (refs. 44,45), are promising candidates.

Conclusions

In summary, we have demonstrated a giant NNE in trilayer graphene at zero magnetic field. The NNE has been further confirmed by the reversal of temperature gradient. Analogous to the scaling law for the NHE, we have established a scaling law for the NNE, revealing the dominant physical origin from skew scattering in the ultrahigh-mobility graphene. Our observation of NNE in graphene will inspire experimental studies in a wide variety of other time-reversal-invariant and non-centrosymmetric materials. To demonstrate this effect at room temperature and modulate it by magnetic fields are promising directions for future studies, with potential to advance its applications in thermoelectric energy harvesting and cooling. Realizing the NNE in three-dimensional bulk materials represents another key direction for future research, which overcomes the limitations of 2D materials.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41565-025-01963-8.

References

1. Xiao, D., Yao, Y., Fang, Z. & Niu, Q. Berry-phase effect in anomalous thermoelectric transport. *Phys. Rev. Lett.* **97**, 026603 (2006).
2. Uchida, K. I. & Heremans, J. P. Thermoelectrics: from longitudinal to transverse. *Joule* **6**, 2240–2245 (2022).
3. Uchida, K. I., Zhou, W. & Sakuraba, Y. Transverse thermoelectric generation using magnetic materials. *Appl. Phys. Lett.* **118**, 140504 (2021).
4. Sakai, A. et al. Giant anomalous Nernst effect and quantum-critical scaling in a ferromagnetic semimetal. *Nat. Phys.* **14**, 1119–1124 (2018).
5. Sakai, A. et al. Iron-based binary ferromagnets for transverse thermoelectric conversion. *Nature* **581**, 53–57 (2020).
6. Pan, Y. et al. Giant anomalous Nernst signal in the antiferromagnet $YbMnBi_2$. *Nat. Mater.* **21**, 203–209 (2022).
7. Asaba, T., Ivanov, V., Thomas, S. M., Savrasov, S. Y. & Ronning, F. Colossal anomalous Nernst effect in a correlated noncentrosymmetric Kagome ferromagnet. *Sci. Adv.* **7**, eabf1467 (2021).
8. Guin, S. N. et al. Zero-field Nernst effect in a ferromagnetic Kagome-lattice Weyl-semimetal $Co_3Sn_2S_2$. *Adv. Mater.* **31**, 1806622 (2019).
9. Yang, H. et al. Giant anomalous Nernst effect in the magnetic Weyl semimetal $Co_3Sn_2S_2$. *Phys. Rev. Mater.* **4**, 024202 (2020).
10. Li, M. et al. Large anomalous Nernst effects at room temperature in Fe_3Pt thin films. *Adv. Mater.* **35**, 2301339 (2023).
11. Miura, A. et al. Observation of anomalous Ettingshausen effect and large transverse thermoelectric conductivity in permanent magnets. *Appl. Phys. Lett.* **115**, 222403 (2019).
12. Miura, A. et al. High-temperature dependence of anomalous Ettingshausen effect in $SmCo_5$-type permanent magnets. *Appl. Phys. Lett.* **117**, 082408 (2020).
13. Zhou, W. et al. Seebeck-driven transverse thermoelectric generation. *Nat. Mater.* **20**, 463–467 (2021).
14. Sodemann, I. & Fu, L. Quantum nonlinear Hall effect induced by Berry curvature dipole in time-reversal invariant materials. *Phys. Rev. Lett.* **115**, 216806 (2015).
15. Ma, Q. et al. Observation of the nonlinear Hall effect under time-reversal-symmetric conditions. *Nature* **565**, 337–342 (2019).
16. Kumar, D. et al. Room-temperature nonlinear Hall effect and wireless radiofrequency rectification in Weyl semimetal $TaIrTe_4$. *Nat. Nanotechnol.* **16**, 421–425 (2021).
17. He, P. et al. Graphene moiré superlattices with giant quantum nonlinearity of chiral Bloch electrons. *Nat. Nanotechnol.* **17**, 378–383 (2022).
18. Kang, K., Li, T., Sohn, E., Shan, J. & Mak, K. F. Nonlinear anomalous Hall effect in few-layer WTe_2. *Nat. Mater.* **18**, 324–328 (2019).
19. Duan, J. et al. Giant second-order nonlinear Hall effect in twisted bilayer graphene. *Phys. Rev. Lett.* **129**, 186801 (2022).
20. He, P. et al. Quantum frequency doubling in the topological insulator Bi_2Se_3. *Nat. Commun.* **12**, 698 (2021).
21. Du, Z. Z., Wang, C. M., Li, S., Lu, H. Z. & Xie, X. C. Disorder-induced nonlinear Hall effect with time-reversal symmetry. *Nat. Commun.* **10**, 3047 (2019).
22. Ma, D., Arora, A., Vignale, G. & Song, J. C. W. Anomalous skew-scattering nonlinear Hall effect and chiral photocurrents in PT-symmetric antiferromagnets. *Phys. Rev. Lett.* **131**, 076601 (2023).
23. Gao, Y. & Xiao, D. Orbital magnetic quadrupole moment and nonlinear anomalous thermoelectric transport. *Phys. Rev. B* **98**, 060402 (2018).
24. Zeng, C., Nandy, S., Taraphder, A. & Tewari, S. Nonlinear Nernst effect in bilayer WTe_2. *Phys. Rev. B* **100**, 245102 (2019).
25. Yu, X.-Q., Zhu, Z.-G., You, J.-S., Low, T. & Su, G. Topological nonlinear anomalous Nernst effect in strained transition metal dichalcogenides. *Phys. Rev. B* **99**, 201410 (2019).
26. Wu, Y.-L., Zhu, G.-H. & Yu, X.-Q. Nonlinear anomalous Nernst effect in strained graphene induced by trigonal warping. *Phys. Rev. B* **104**, 195427 (2021).
27. Papaj, M. & Fu, L. Enhanced anomalous Nernst effect in disordered Dirac and Weyl materials. *Phys. Rev. B* **103**, 075424 (2021).
28. Ghahari, F. et al. Enhanced thermoelectric power in graphene: violation of the Mott relation by inelastic scattering. *Phys. Rev. Lett.* **116**, 136802 (2016).
29. Small, J. P., Perez, K. M. & Kim, P. Modulation of thermoelectric power of individual carbon nanotubes. *Phys. Rev. Lett.* **91**, 256801 (2003).
30. Duan, J. et al. High thermoelectric power factor in graphene/hBN devices. *Proc. Natl Acad. Sci. USA* **113**, 14272–14276 (2016).
31. Xu, L. et al. Second harmonic generation induced by gate voltage oscillation in few layer $MnBi_2Te_4$. *npj Quantum Mater.* **9**, 79 (2024).
32. Wei, P., Bao, W., Pu, Y., Lau, C. N. & Shi, J. Anomalous thermoelectric transport of Dirac particles in graphene. *Phys. Rev. Lett.* **102**, 166808 (2009).
33. Zuev, Y. M., Chang, W. & Kim, P. Thermoelectric and magneto-thermoelectric transport measurements of graphene. *Phys. Rev. Lett.* **102**, 096807 (2009).
34. Zeng, C., Nandy, S. & Tewari, S. Fundamental relations for anomalous thermoelectric transport coefficients in the nonlinear regime. *Phys. Rev. Res.* **2**, 032066 (2020).
35. Zeng, C., Nandy, S. & Tewari, S. Chiral anomaly induced nonlinear Nernst and thermal Hall effects in Weyl semimetals. *Phys. Rev. B* **105**, 125131 (2022).
36. Wang, Y., Zhu, Z.-G. & Su, G. Quantum theory of nonlinear thermal response. *Phys. Rev. B* **106**, 035148 (2022).
37. Xiao, C., Du, Z. Z. & Niu, Q. Theory of nonlinear Hall effects: modified semiclassics from quantum kinetics. *Phys. Rev. B* **100**, 165422 (2019).
38. Mehraeen, M. Quantum kinetic theory of quadratic responses. *Phys. Rev. B* **110**, 174423 (2024).

39. Isobe, H., Xu, S.-Y. & Fu, L. High-frequency rectification via chiral Bloch electrons. *Sci. Adv.* **6**, eaay2497 (2020).
40. Makushko, P. et al. A tunable room-temperature nonlinear Hall effect in elemental bismuth thin films. *Nat. Electron.* **7**, 207–215 (2024).
41. Choi, S. H. et al. Large-scale synthesis of graphene and other 2D materials towards industrialization. *Nat. Commun.* **13**, 1484 (2022).
42. Liu, L. et al. A mass transfer technology for high-density two-dimensional device integration. *Nat. Electron.* **8**, 135–146 (2025).
43. Min, L. et al. Strong room-temperature bulk nonlinear Hall effect in a spin-valley locked Dirac material. *Nat. Commun.* **14**, 364 (2023).
44. Cheng, B. et al. Giant nonlinear Hall and wireless rectification effects at room temperature in the elemental semiconductor tellurium. *Nat. Commun.* **15**, 5513 (2024).
45. Liu, H. et al. Giant nonlinear Hall effect induced ultrahigh rectification in a Weyl semiconductor. *Adv. Electron. Mater.* **11**, 2400648 (2024).

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© The Author(s), under exclusive licence to Springer Nature Limited 2025

Methods

Device fabrication

The h-BN–graphene–h-BN heterostructures were fabricated using a dry-transfer technique. h-BN (HQ Graphene) and graphene (HOPG, HQ Graphene) were first mechanically exfoliated on separate Si/SiO_2 (285 nm) substrates. Next, suitable h-BN thin films (20–40 nm thick) and trilayer graphene flakes were identified through optical microscopy and Raman spectroscopy. Then, the dry-transfer process via polydimethylsiloxane stamping was used to pick up the top layer of h-BN. Subsequently, the graphene layer was picked by the top h-BN using the same method while maximizing the interface area. Finally, the stack was deposited onto a bottom h-BN flake, creating the final heterostructure. To fabricate the Hall bar device with additional heaters and thermometers, a two-step process with electron beam lithography and reactive ion etching was utilized. The electron beam resist MMA EL6 and PMMA A4 were spin-coated onto the sample for etch mask writing. After etching and cleaning the resist off, electron beam lithography was used to define the electrodes on a newly spin-coated resist. Cr and Au (3 nm and 90 nm, respectively) metal layers were deposited on the written areas by using a thermal evaporator, creating the final device. After device fabrication, we verified that the graphene layer and Si substrate are electrically isolated by the dielectric layers (with a resistance of ~10^{10} Ω), ensuring effective control of the Fermi level via back-gate voltage. In addition, we confirmed that the thermometers are electrically isolated from the graphene layer (with a resistance of ~10^{11} Ω) owing to the intervening h-BN layer, achieved through careful design.

Thermoelectrical and electrical measurement

The thermoelectrical and all-electric measurements were performed in a Physical Properties Measurement System (PPMS, Quantum Design). All measurements were carried out at zero magnetic field except for the carrier density determination. A sinusoidal current $I_\omega = I\sin\omega t$ with a frequency of $\omega/2\pi = 17.777$ Hz was applied to the heater for the thermoelectrical measurement or to the Hall device for the electrical measurement using a source meter (Keithley 6221). The different harmonic voltages along the longitudinal and transverse directions were measured by lock-in amplifiers (Stanford Research SR830). The phases of lock-in amplifiers were set to 0° and 90° for the odd and even harmonic voltage measurements, respectively. A subfemtoamp source meter (Keithley 6430) was used for the back-gate voltage application and leakage current measurement. Two Keithley 2400 source meters were used to apply d.c. probing currents, and the four-probe resistance of two thermometers was measured simultaneously using lock-in amplifiers. The bonding wires used in our measurements are made of aluminium, whose Seebeck coefficient is negligibly small. This ensures that the measured thermoelectric signals originate primarily from the graphene layer.

Determination of temperature variations at the two thermometers

The temperature variation δT at the thermometer is obtained by probing the change in its resistance ΔR. When an a.c. current I_h^ω with frequency of ω is applied to the heater, the temperature variation at the thermometer fluctuates with a frequency of 2ω due to the Joule heating, $\delta T^{2\omega} \propto (I_h^\omega)^2$. Thus, the ΔR would also fluctuate with 2ω, as $\Delta R \propto \delta T^{2\omega}$ at a small δT. This is because the temperature-dependent resistance R can be expressed as a function of δT as $R = R_0 + \frac{dR}{dT}\delta T + \ldots$, according to the Taylor expansion. When a d.c. current $I_0^{d.c.}$ is further applied to the thermometer, the voltage drop along the thermometer can be expressed as

$$V = I_0^{d.c.}(R_0 + \Delta R) = I_0^{d.c.}R_0 + I_0^{d.c.}\frac{dR}{dT}\delta T^{2\omega} = V_0^{d.c.} + V^{2\omega}.$$

We find the δT is related with the second-harmonic voltage $V^{2\omega} = I_0^{d.c.}\frac{dR}{dT}\delta T^{2\omega}$, which can be measured using a lock-in amplifier. Thus, the δT at the thermometer can be determined as

$$\delta T = \frac{V^{2\omega}}{I_0^{d.c.}}\left(\frac{dR}{dT}\right)^{-1}.$$

Theoretical NNE conductivity

Driven by the temperature gradient ∇T, the current of NNE can be written as[1] ($\hbar = 1$)

$$J_a = -e\int\frac{d^2k}{(2\pi)^2}(v_{0,a} + v_{sj,a})F(r,k) - e\epsilon_{abc}\frac{\nabla T_b}{T}\int\frac{d^2k}{(2\pi)^2}\Omega_c(k),$$
$$[(\varepsilon_k - \mu)]F(r,k) = \alpha_{abc}(-\nabla T_b)(-\nabla T_c)$$

where $e(e<0)$ is the charge on the electron, k is the wave vector, $v_0 = \frac{\partial\varepsilon_k}{\partial k}$ is the group velocity of electrons, v_{sj} is the side-jump velocity, $F(r,k)$ is the Fermi–Dirac distribution function, ϵ_{abc} is the Levi–Civita symbol, $\Omega(k)$ is the Berry curvature, ε_k is the eigen energy, μ is the chemical potential and $a,b,c = x,y$. The NNE conductivity induced by the intrinsic contribution can be written as[24,25]

$$\alpha_{abc}^{int} = -e\tau\epsilon_{abd}\int\frac{d^2k}{(2\pi)^2}\frac{(\varepsilon_k - \mu)^2}{T^2}\Omega_d(k)\frac{\partial\varepsilon_k}{\partial k_c}\frac{\partial F^{(0)}(r,k)}{\partial\varepsilon_k},$$

where $F^{(0)}(r,k)$ is the equilibrium distribution function (see more details in Supplementary Note 3). This conductivity is proportional to relaxation time τ, and we label it as intrinsic term α_{abc}^{int} because no disorder scattering is involved. Importantly, the maximum symmetry for a 2D system that allows α_{abc}^{int} to exist is a single mirror line, which is not met in ABA graphene with a C_3 symmetry[25]. Thus, it is reasonable to speculate that the second-order contribution in our work is due to impurity scattering.

When considering side-jump (sj) and skew (sk) scattering, NNE conductivity arising from side-jump reads[39]

$$\alpha_{abc}^{sj} = e\tau^2\int\frac{d^2k}{(2\pi)^2}(v_{0,a}v_{0,b}v_{c,sj} - v_{a,sj}v_{0,b}v_{0,c})$$
$$\left[\frac{(\varepsilon_k - \mu)^2}{T^2}\frac{\partial^2 f_0}{\partial\varepsilon_k^2} + \frac{2}{T}\frac{(\varepsilon_k - \mu)}{T}\frac{\partial f_0}{\partial\varepsilon_k}\right],$$

and the skew scattering term is

$$\alpha_{abc}^{sk} = -\frac{e\tau^3}{T^2}\int_k v_{0,a}\int_{k'}\frac{\partial\varepsilon_k}{\partial k'_b}\frac{\partial\varepsilon_k}{\partial k'_c}w_{kk'}^{(A)}\left[4(\varepsilon_k - \mu)\frac{\partial f_0}{\partial\varepsilon_k} + 2(\varepsilon_k - \mu)^2\frac{\partial^2 f_0}{\partial\varepsilon_k^2}\right]$$

where $w_{kk'}^{(A)}$ is the anti-symmetric scattering rate and f_0 is the equilibrium Fermi–Dirac distribution function. Constrained by the C_3 symmetry, the independent elements of side-jump and skew scattering are $\alpha_{xyy}^{sj/sk}$ and $\alpha_{yxx}^{sj/sk}$. Furthermore, we note that the two commonly considered side-jump contributions from the side-jump velocity and the anomalous distribution cancel each other out (Supplementary Note 3.3).

Data availability

The data that support the findings of this study are available within the Article and its Supplementary Information. Other relevant data are available from the corresponding authors upon reasonable request. Source data are provided with this paper.

Code availability

The codes that support this study are available from the corresponding author upon reasonable request.

Acknowledgements
We acknowledge Y. Zhu and Y. Zhou for helpful discussions. P.H. was sponsored by National Key Research and Development Program of China (grant nos. 2022YFA1403300 and 2020YFA0308800), National Natural Science Foundation of China (grant nos. 12174063 and U23A2071), Natural Science Foundation of Shanghai (grant no. 23ZR1403600) and the start-up funding from Fudan University. H.J. was sponsored by National Key Research and Development Program of China (grant no. 2022YFA1403700) and National Natural Science Foundation of China (grant no. 12350401). This work was supported by Innovation Program for Quantum Science and Technology grant no. 2024ZD0300103.

Author contributions
P.H. and J.S. planed the study. H.L. and P.H. performed the transport measurements and the data analysis. J.L., H.L., J.Z. and M.Z. fabricated devices. Z.Z., H.J. and X.C.X. performed theoretical studies. P.H., H.L., Z.Z., H.J., X.C.X. and J.S. wrote the paper. All authors commented on the paper.

Competing interests
The authors declare no competing interests.

Additional information
Extended data is available for this paper at https://doi.org/10.1038/s41565-025-01963-8.

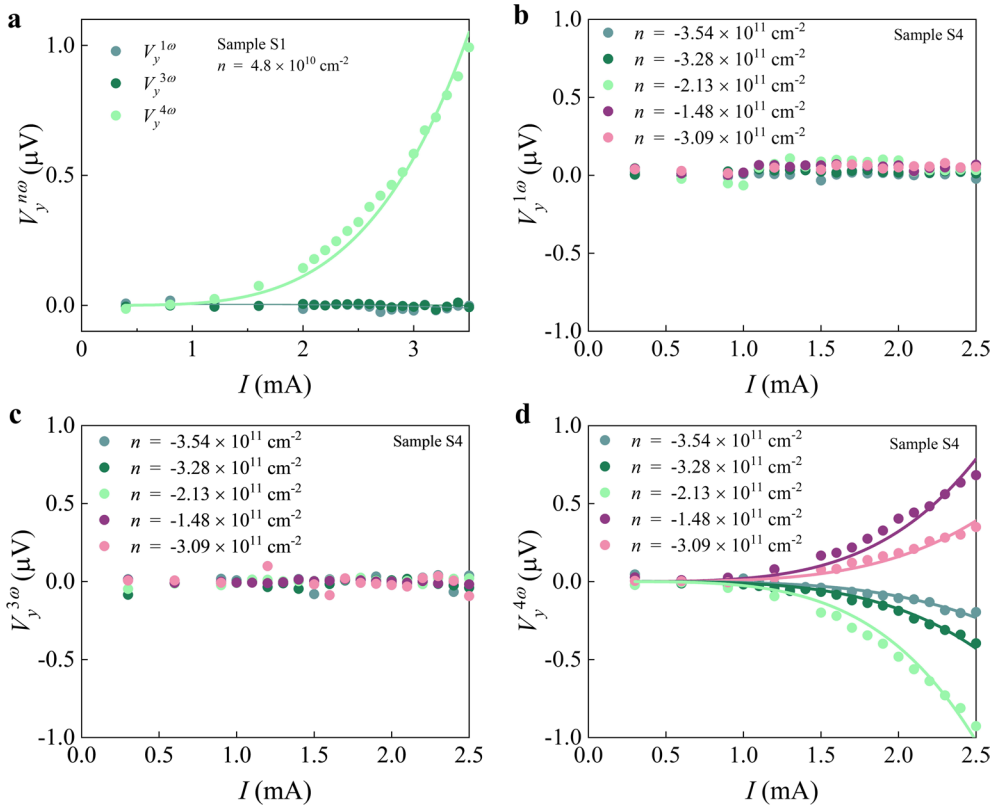
Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41565-025-01963-8.

Correspondence and requests for materials should be addressed to Hua Jiang, Pan He or Jian Shen.

Peer review information *Nature Nanotechnology* thanks the anonymous reviewers for their contribution to the peer review of this work.

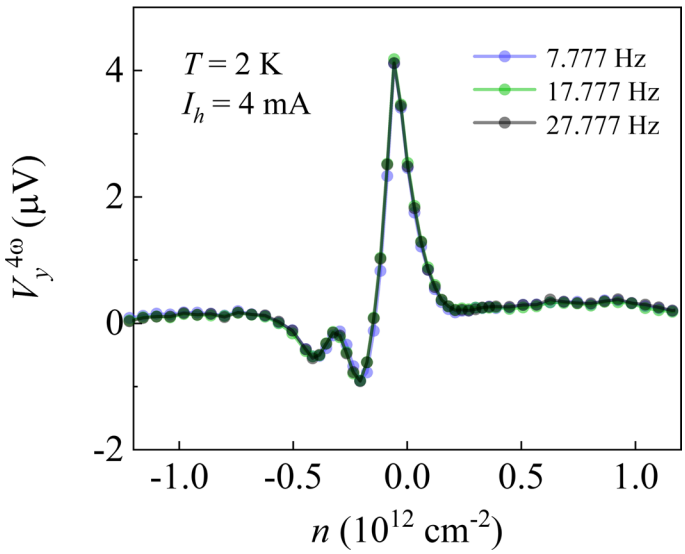
Reprints and permissions information is available at www.nature.com/reprints.

https://doi.org/10.1038/s41565-025-01963-8



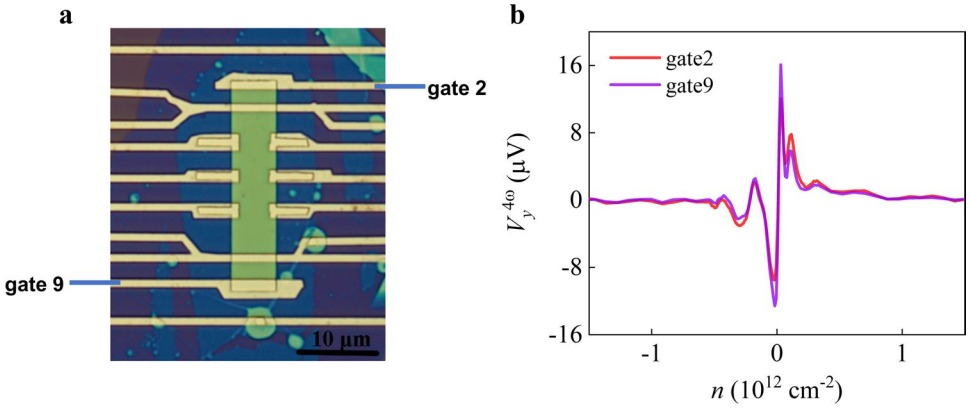
Extended Data Fig. 1 | Simultaneously measured 1ω, 3ω, and 4ω transverse voltages. a, Simultaneously measured 1ω, 3ω, and 4ω transverse voltages as a function of the heating current for a representative carrier density in Sample S1. **b**-**d**, Simultaneously measured 1ω (**b**), 3ω (**c**), and 4ω (**d**) transverse voltages as a function of the heating current at different carrier densities in Sample S4. The 1ω and 3ω transverse voltages remain negligible within the noise level in both samples, in stark contrast to the notable 4ω voltage.

https://doi.org/10.1038/s41565-025-01963-8



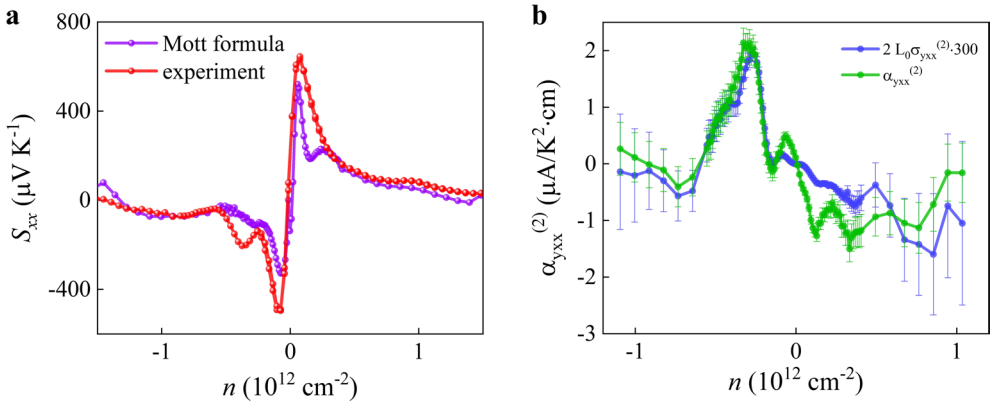
Extended Data Fig. 2 | Frequency-independent nonlinear Nernst effect. The fourth harmonic voltage $V_y^{4\omega}$ measured at three different frequencies in sample S1. The amplitude of heating current is 4 mA. The $V_y^{4\omega}$ stays the same for all three frequencies, consistent with the nature of NNE.

https://doi.org/10.1038/s41565-025-01963-8



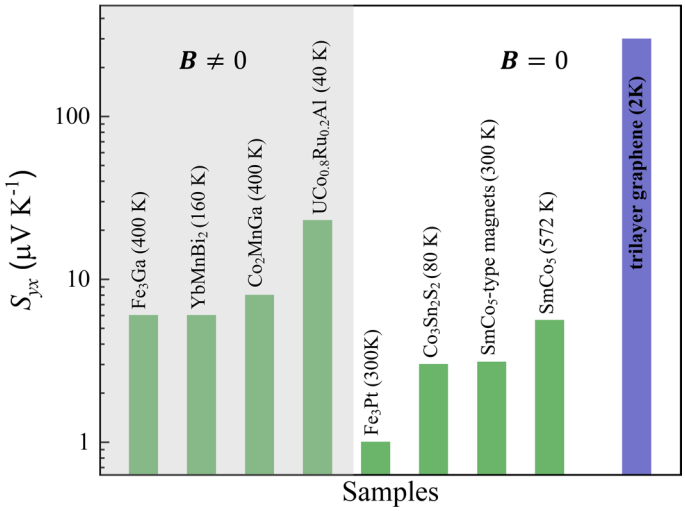
Extended Data Fig. 3 | Excluding the fourth harmonic signals from gate voltage oscillations. a, An optical image of device with two gate electrodes on both sides as marked in sample S1. One of them is connected to the negative pole of gate voltage source. The positive pole of gate voltage source is always connected to the highly-doped silicon substrate. **b**, The fourth harmonic voltage $V_y^{4\omega}$ as a function of carrier density. The data were measured when the negative pole of gate voltage is connected to either electrode 2 or 9 at 2 K. The $V_y^{4\omega}$ is almost the same for the two different cases. This excludes the possibility of $V_y^{4\omega}$ from the gate effect[31].

https://doi.org/10.1038/s41565-025-01963-8



Extended Data Fig. 4 | Mott relation in both linear and nonlinear regime in sample S1. The linear Seebeck coefficient S_{xx} and nonlinear Nernst conductivity $\alpha_{yxx}^{(2)}$ as a function of carrier density at 5 K in sample S1. **a**, Comparison between the measured S_{xx} (red line) and the one calculated based on Mott formula (purple line) using the gate-dependent conductance at $T = 5$ K. **b**, The experimental $\alpha_{yxx}^{(2)}$ and the calculated $\alpha_{yxx}^{(2)}$ from the Mott formula using the measured $\sigma_{yxx}^{(2)}$ are plotted together for a direct comparison. The Mott formula based on BCD-like theory[34–36] states that the nonlinear Nernst conductivity $\alpha_{yxx}^{(2)}$ is proportional to the nonlinear Hall conductivity $\alpha_{yxx}^{(2)}$ with $\alpha_{yxx}^{(2)} = 2L_0\sigma_{yxx}^{(2)}$, where $L_0 = \frac{k_B^2\pi^2}{3e^2}$ is the Lorentz number. A similar carrier density dependence is found for the measured $\alpha_{yxx}^{(2)}$ and the calculated one. They show a 300 times difference in magnitude. The error bars represent the standard deviation of the nonlinear conductivity. The error bar is the standard error of calculated nonlinear conductivity raised from the voltage noise of 0.05 μV.

https://doi.org/10.1038/s41565-025-01963-8



Extended Data Fig. 5 | Summary of Nernst coefficient from different materials. Absolute value of anomalous Nernst coefficient (S_{yx}) from ferromagnets Fe_3Ga, Co_2MnGa, $UCo_{0.8}Ru_{0.2}Al$, Fe_3Pt, $Co_3Sn_2S_2$, $SmCo_5$, $SmCo_5$-type and antiferromagnets $YbMnBi_2$ are obtained from literature[4–12]. The Nernst coefficient measured in this work from ABA trilayer graphene is shown as a violate bar for comparison. The results in the gray box were measured with a magnetic field, while the results outside the box were measured at zero magnetic field. The vertical axis is on a logarithmic scale.